

Project Saransh: Technical Stack & Architecture

Complete Engineering Blueprint: Frontend, Backend, AI Pipeline & ABDM Interoperability

Frontend Runtime: React 18 + Vite 8.3 (Single Page App, 60 FPS, Offline IndexedDB)
Backend Runtime: FastAPI (Python 3.14) + Uvicorn ASGI Server (Async, Sub-20ms P99 Latency)
AI & Safety Core: Google Gemini 2.5 Flash + Deterministic Non-AI Python Guardrails
National Standards: ABDM (M1, M2, M3), HL7 FHIR Release 4, SNOMED-CT, ICD-10, LOINC, DPDP Act 2023

1. System Architecture & High-Level Topology

Saransh is structured around a **decoupled, offline-first client-server architecture** designed to operate reliably across varying connectivity tiers in Indian government health facilities. The frontend runs as a client-side Progressive Web App on low-cost Android tablets or desktop kiosks, backed by an asynchronous FastAPI microservice capable of deploying on local edge servers (PHC LAN) or state cloud nodes (State Data Centres / MeghRaj).

2. Frontend Technology Stack (Client & Edge Workstation)

Component / Library	Version / Specification	Role, Function & Architectural Value
React	18.3.1 (Concurrent)	Component-based reactive UI engine with Concurrent Mode rendering, state isolation, and zero layout shift during high-speed triage intake.
Vite	8.3.1 (Rollup/ESBuild)	Modern frontend build tool delivering lightning-fast Hot Module Replacement (HMR) and optimized production bundles built in under 1.2 seconds.
Tailwind CSS	3.4+ (PostCSS)	Utility-first styling enforcing a strict 4px/8px spacing grid, MedTech Obsidian dark/light themes, and WCAG AA/AAA contrast ratios for outdoor clinics.
Lucide React	Latest (Tree-shaken)	Clean, standardized clinical and operational iconography (Stethoscope, ShieldCheck, Heart, Thermometer, AlertTriangle, Building2).
Web Speech API	Native Browser Spec	Direct client-side acoustic speech recognition and voice recording without requiring external proprietary audio streaming SDKs.
HTML5 SVG Canvas	Vector Topological	Interactive 2D anatomical human body map with 6 distinct clinical zones (Head, Chest, Lungs, Abdomen, Limbs, Skin) and Front/Back perspective toggles.
IndexedDB Client Cache	Browser Local Storage API	Asynchronous offline queue enabling frontline workers to register patients, log vitals, and queue records locally during broadband outages.

3. Backend Technology Stack (FastAPI & Server Pipeline)

Component / Library	Version / Specification	Role, Function & Architectural Value
FastAPI	0.115+ (Python 3.14)	High-performance asynchronous web framework providing automated OpenAPI/Swagger documentation, native async coroutines, and sub-20ms endpoint latency.
Uvicorn	0.34+ (ASGI Server)	Lightning-fast ASGI server implementation based on uvloop and httptools, capable of sustaining thousands of concurrent health worker polling connections.
Pydantic v2	2.10+ (Rust Core)	Strict data validation and serialization across 10 clinical data schemas: PatientInfo, Symptoms, Vitals, MedicalHistory, TriageOutput, FHIR R4 Bundle.
SQLite / SQLAlchemy	3.45+ (ACID Compliant)	Zero-configuration, serverless embedded relational database engine ideal for rural edge deployments. Supports seamless migration to PostgreSQL for state servers.
Cryptographic Ledger	Standard Python hashlib	SHA-256 tamper-evident audit hashing for doctor triage overrides, NHM referral slips, and clinical counter-signatures.

4. AI Engine, Indic NLP & Deterministic Safety Gatekeeper

Intelligence Layer	Technology / Model	Function & Mechanism
Deterministic Red-Line Engine	Pure Python Rule Engine (Zero AI / No Hallucination)	Hard physiological guardrails evaluated before and after any AI inference. If SpO2 < 90%, SBP > 180, or severe crushing chest pain is detected, the patient is locked into the ■ RED lane immediately.
Clinical LLM Normalizer	Google Gemini 2.5 Flash API (google-genai SDK)	Synthesizes multimodal complaints, extracts onset/duration, maps unstructured vernacular colloquialisms into formal English clinical summaries, and suggests differential triage lanes.
Indic Speech Normalizer	Custom Linguistic Map (Odia / Hindi)	Translates cultural idioms (e.g. Odia 'Chhati re bhara') directly into standardized medical ontologies: SNOMED-CT (225566008) and ICD-10 (I20.9) .

5. National Healthcare Standards & Interoperability

Standard / Framework	Governing Authority	Implementation & Compliance In Saransh
ABDM Milestones (M1, M2, M3)	National Health Authority (NHA)	M1: 14-digit ABHA ID validation & demographic prefill; M2: Standard FHIR R4 Bundle creation; M3: Health Information Exchange gateway readiness.
HL7 FHIR Release 4	HL7 International & MoHFW	Automated export of Condition, Observation, Encounter, and Patient resources formatted in standardized JSON bundles for hospital HMIS sharing.
SNOMED-CT	SNOMED International	Granular clinical terminology mapping for symptoms, anatomical sites, and clinical findings (e.g. 225566008 for Chest Pain / Angina).
ICD-10	World Health Organization	International Classification of Diseases codes for provisional diagnostic categories (e.g. I20.9 for Angina, R06.02 for Dyspnea, R42 for Vertigo).
NHM Referral Standard	National Health Mission (NHM)	Formal counter-signed inter-facility referral slip generation with facility codes (PHC Jatni -> Capital Hospital / SCB Cuttack), ambulance checklist, and doctor signature hash.

6. Security, Privacy & Regulatory Governance Stack

Governance Pillar	Regulatory Framework	Technical Safeguards Enforced
DPDP Act 2023	MeitY, Government of India	Client-side PII stripping before LLM calls; Section 7(a) emergency casualty exemption for unconscious trauma casualties; explicit consent capture.
CDSCO SaMD Class A	CDSCO Medical Device Rules 2017	Classified as an Assistive Clinical Decision Support System (CDSS Class A). Software does not independently prescribe or formulate diagnoses.
NMC Physician Primacy	National Medical Commission	100% legal diagnosis remains with the registered Medical Officer. Mandatory counter-signature with State/NMC Registration Number.
Data Encryption	National Cyber Security Norms	AES-GCM-256 encryption for local IndexedDB and SQLite storage at rest; TLS 1.3 enforced for all network transit.

Deployment Hardware Specifications:

- **Client Workstations:** Any standard Android tablet ($\geq$ Android 10, 3GB RAM) or desktop browser (Chrome/Edge/Firefox).
- **Edge Server:** Raspberry Pi 4 / Intel Celeron N100 mini-PC / Local Clinic Desktop running Python 3.12+ and SQLite.
- **Cloud Tier:** Scalable Linux container (Docker) deployable to NIC MeghRaj Cloud, AWS Mumbai, or State Data Centres.